Worked Solutions to Chapter 5: Chaos in Two-Dimensional Maps

Chaos: An Introduction to Dynamical Systems (Alligood, Sauer, Yorke)

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This document solves the end-of-chapter Exercises 5.1–5.10 and Challenge 5 of Chapter 5. Each solution gives the main idea, then the steps, then a boxed answer. Problem statements are paraphrased, so keep the book open next to this text. Final answers were checked against the book's "Answers and Hints" appendix wherever it gives one.

Toolkit: facts used throughout

Toolkit

1. **Lyapunov numbers.** Let $J_n = Df^n(v_0)$ and let $r_1^n \geq \cdots \geq r_m^n$ be the lengths of the axes of the ellipsoid J_n(unit sphere). These axes are the singular values of J_n (square roots of the eigenvalues of $J_n J_n^T$), so $r_1^n \cdots r_m^n = |\det J_n|$. Then $L_k = \lim_n (r_k^n)^{1/n}$ and the exponents are $h_k = \ln L_k$. If J_n is a power A^n of a symmetric matrix A, the axes are $|\lambda_k|^n$ for the eigenvalues λ_k of A.
2. **Chaotic orbit (Def. 5.2).** A bounded orbit that is not asymptotically periodic, has no Lyapunov number equal to 1, and has $L_1 > 1$ (i.e. $h_1 > 0$).
3. **Fixed-point tools.** *Theorem 5.10:* if f is continuous and the vectors $f(x) - x$ make a nonzero net number of turns as x goes once around the boundary of a rectangle S, then f has a fixed point in S. *Corollary 5.13:* if $f(S_i)$ "lies across" S_{i+1} for $i = 1, \ldots, k-1$ and $f(S_k)$ lies across S_1, all with the same orientation, then f^k has a fixed point in S_1 whose orbit visits $S_1, \ldots, S_k$.
4. **Skinny baker map** $B(x_1, x_2) = (\frac{1}{3}x_1, 2x_2)$ for $x_2 \leq \frac{1}{2}$, and $(\frac{1}{3}x_1 + \frac{2}{3}, 2x_2 - 1)$ for $x_2 > \frac{1}{2}$. Every orbit has Lyapunov exponents $\ln 2$ and $-\ln 3$. Points of the invariant set have two-sided itineraries over $\{L, R\}$; B acts as the shift. An orbit is asymptotically periodic iff its itinerary is eventually periodic toward the right.
5. **Horseshoe map** h (with uniform stretching and contraction): points of the invariant set H correspond one-to-one to two-sided L/R itineraries, h acts as the shift, and the Lyapunov exponents are $\ln 3$ and $-\ln 4$ for the book's model.

Exercises

Exercise 5.1

If $\det Df(x) = D$ *for every* x*, explain why the product of all* m *Lyapunov numbers is* $|D|$ *(the exponents add up to* $\ln|D|$*).*

Here $J_n = Df(v_{n-1})\cdots Df(v_0)$, so $\det J_n = D^n$. The product of the semi-axes of the ellipsoid $J_n(\text{sphere})$ equals $|\det J_n| = |D|^n$, so $r_1^n \cdots r_m^n = |D|^n$. Take nth roots:

$$(r_1^n)^{1/n} \cdots (r_m^n)^{1/n} = |D| \quad \text{for every } n.$$

Let $n \to \infty$ (assuming the Lyapunov numbers exist): $L_1 L_2 \cdots L_m = |D|$. Taking logarithms, $h_1 + \cdots + h_m = \ln|D|$. □

Answer

The product of the semi-axes of $J_n(\text{sphere})$ is $|\det J_n| = |D|^n$; taking nth roots and limits gives $\prod L_k = |D|$ and $\sum h_k = \ln|D|$ (this is D itself when $D > 0$).

Exercise 5.2

Find the Lyapunov exponents of $f(\mathbf{x}) = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix} \mathbf{x} \pmod 1$ *and explain why they are half those of the cat map.*

The Jacobian is the constant symmetric matrix $M = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$. Its eigenvalues solve $\lambda^2 - \lambda - 1 = 0$:

$$\lambda_{1,2} = \frac{1 \pm \sqrt{5}}{2} \approx 1.618,\ -0.618.$$

Since M is symmetric, the axes of $M^n(\text{sphere})$ are $|\lambda_k|^n$. So the Lyapunov numbers are $\frac{1+\sqrt{5}}{2}$ and $\frac{\sqrt{5}-1}{2}$ (the same for every orbit, since Df is constant), and

$$h_1 = \ln\frac{1+\sqrt{5}}{2} \approx 0.481, \qquad h_2 = \ln\frac{\sqrt{5}-1}{2} \approx -0.481.$$

Why half of the cat map? Compute $M^2 = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix} = A$, the cat-map matrix. Reducing mod 1 commutes with integer matrices, so $f \circ f$ *is* the cat map. Applying a map twice squares the expansion factors, so it doubles the exponents. Hence the cat map's exponents (± 0.962) are twice these.

Answer

$h_1 = \ln\frac{1+\sqrt{5}}{2} \approx 0.481$ and $h_2 = \ln\frac{\sqrt{5}-1}{2} \approx -0.481$. The cat-map matrix equals M^2, so its exponents are doubled.

Exercise 5.3

Show that the cat map $f(\mathbf{v}) = A\mathbf{v} \pmod 1$, $A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$, *is one-to-one.*

Suppose $A\mathbf{v} \equiv A\mathbf{w} \pmod 1$ for $\mathbf{v}, \mathbf{w} \in [0,1)^2$. Then $A(\mathbf{v} - \mathbf{w}) = \mathbf{n}$ for some integer vector $\mathbf{n} \in \mathbb{Z}^2$. Since $\det A = 1$, the inverse $A^{-1} = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}$ has integer entries, so $\mathbf{v} - \mathbf{w} = A^{-1}\mathbf{n} \in \mathbb{Z}^2$. Each coordinate of $\mathbf{v} - \mathbf{w}$ lies in $(-1, 1)$, and the only integer in $(-1, 1)$ is 0. So $\mathbf{v} = \mathbf{w}$. □

Answer

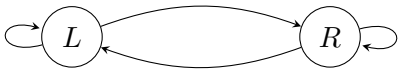
$A\mathbf{v} \equiv A\mathbf{w} \pmod 1$ forces $\mathbf{v} - \mathbf{w} = A^{-1}\mathbf{n}$ to be an integer vector (because A^{-1} has integer entries), and that vector must be $\mathbf{0}$.

Exercise 5.4

Draw the transition graphs for the Markov partitions of the skinny baker map and the horseshoe map.

Baker map. The partition is $\{L, R\}$, the left and right halves of the square. $B(L)$ consists of two thin vertical strips (one in each half) and each strip lies across L or R vertically; the same holds for $B(R)$. So every transition $L \to L$, $L \to R$, $R \to L$, $R \to R$ is allowed.

Horseshoe. The partition is $\{V_L, V_R\}$, the two vertical legs of $W \cap h(W)$. The horizontal strips $\cdot L$ and $\cdot R$ of W are mapped across V_L and V_R respectively, and each leg contains pieces of both strips. So $h(V_L)$ and $h(V_R)$ each lie across both V_L and V_R: all four transitions are again allowed.



Answer

Both maps have the same transition graph: two vertices (L and R) with arrows $L \to L$, $L \to R$, $R \to L$, $R \to R$. Every two-sided L/R sequence is allowed (the "full 2-shift").

Exercise 5.5

Show that the set of chaotic orbits of the horseshoe map is uncountable.

Use the itinerary description. Points of the invariant set H correspond one-to-one to two-sided sequences over $\{L, R\}$. An orbit is asymptotically periodic exactly when its itinerary is eventually periodic toward the right; all orbits in H are bounded and have exponents $\ln 3 > 0$ and $-\ln 4$ (neither equal to 0). So *an orbit in H is chaotic iff its right-hand itinerary is not eventually periodic.*

Counting. The set of one-sided sequences $S_1 S_2 S_3 \ldots$ over $\{L, R\}$ is uncountable (Cantor's diagonal argument: they are binary expansions of reals). The eventually periodic ones are countable: such a sequence is determined by finite data (a preperiod block and a period block), and there are

countably many finite words. Hence uncountably many right-hand sequences are *not* eventually periodic. Fix the left side to be $\dots LLL$. Different right-hand sequences give different points of H (one-to-one correspondence), so uncountably many points of H have chaotic orbits. Each orbit is a countable set, and an uncountable set cannot be covered by countably many countable sets, so there are uncountably many distinct chaotic *orbits*. □

Answer

Chaotic orbits $\leftrightarrow$ itineraries that are not eventually periodic to the right. There are uncountably many such sequences (all sequences minus a countable set), hence uncountably many chaotic orbits.

Exercise 5.6

Show that the invariant set H of the horseshoe map contains a dense chaotic orbit.

Idea: build an itinerary that contains every finite block of symbols, as for the dense orbit of the logistic map $G(x) = 4x(1-x)$.

Construction. List all finite words over $\{L, R\}$ in order of length: $w_1, w_2, \dots = L,\ R,\ LL,\ LR,\ RL,\ RR,\ LLL, \dots$. Define the two-sided itinerary

$$\cdots w_3\, w_2\, w_1 \ \bullet\ w_1\, w_2\, w_3 \cdots$$

(the same list on the left, so that backward orbits are dense too). Let $v \in H$ be the point with this itinerary.

Dense. Let $z \in H$ and $\varepsilon > 0$. The set of points of H sharing the symbols $S_{-k} \dots S_k$ of z lies in a small rectangle whose diameter tends to 0 as $k \to \infty$ (the strips have width and height $\sim 4^{-k}$ by the uniform contraction and expansion). Choose k so that its diameter is below ε. The block $S_{-k} \dots S_k$ is some word w_j, and it occurs somewhere to the right of the dot, say starting at position $n-k$. Shifting by n puts that block around the dot, so $h^n(v)$ lies in the same small rectangle as z, i.e. $|h^n(v) - z| < \varepsilon$. So the orbit of v is dense in H.

Chaotic. The itinerary contains arbitrarily long runs $LL \cdots L$, so it is not eventually periodic toward the right, and by Exercise 5.5 the orbit is chaotic. □

Answer

The orbit of the point with itinerary $\cdots w_2 w_1 \bullet w_1 w_2 \cdots$ (all finite words listed by length) visits every small rectangle of H, so it is dense. It is not eventually periodic, hence chaotic.

Exercise 5.7

For $f(z) = z^2$ (z complex): (a) write f in polar coordinates; (b) find the bounded orbits not converging to 0; (c) find their Lyapunov exponents; (d) show f has chaotic orbits.

(a) Write $z = re^{i\theta}$. Then $z^2 = r^2 e^{2i\theta}$, so $p(r, \theta) = (r^2, 2\theta)$.

(b) After n steps $r_n = r^{2^n}$. If $r > 1$, then $r_n \to \infty$. If $r < 1$, then $r_n \to 0$. If $r = 1$, then $r_n = 1$ forever. So the bounded orbits that do not tend to 0 are exactly the points of the **unit circle** $r = 1$.

(c) The derivative of $z \mapsto z^2$ is multiplication by $2z$, a rotation combined with scaling by $|2z|$ (so the Jacobian is a multiple of a rotation, and all axes of the image ellipse are equal). Along an orbit $z_0, z_1, \ldots$ the total scaling after n steps is $\prod_{j<n} 2|z_j|$.

- On the unit circle, $|z_j| = 1$, so the scaling is 2^n in both directions. Lyapunov numbers $2, 2$, exponents $\ln 2, \ln 2$.
- Inside the unit circle ($0 < r < 1$): the scaling is $2^n r^{1+2+\cdots+2^{n-1}} = 2^n r^{2^n-1}$, whose nth root $2r^{(2^n-1)/n} \to 0$. The Lyapunov numbers are $0, 0$, so the exponents do not exist (they are $-\infty$).

(d) On the circle, $\theta \mapsto 2\theta$ is the doubling map $x \mapsto 2x \pmod 1$ with $\theta = 2\pi x$. Take x irrational. Then the orbit is bounded and never eventually periodic. It is also not asymptotically periodic: under doubling, the distance between two nearby points doubles each step (while small), so an orbit converging to a periodic orbit would have to land on it exactly, which is impossible for irrational x. Both Lyapunov numbers equal 2 ($\neq 1$, and $L_1 = 2 > 1$). So these orbits are chaotic, and there are uncountably many of them.

Answer

(a) $p(r, \theta) = (r^2, 2\theta)$. (b) The unit circle $r = 1$. (c) On the circle, exponents $\ln 2$ and $\ln 2$; inside the circle the Lyapunov numbers are $0, 0$ so the exponents do not exist. (d) Orbits with irrational $\theta/2\pi$ are chaotic.

Exercise 5.8

Explore itineraries and periodic points for the two maps of Figure 5.22: (a) the image of the unit square meets the square in two pieces; (b) in three pieces.

Reading the figure. In each picture the image of the square S is a long thin strip folded back over S. The strip crosses S in N **vertical legs** $V_1, \ldots, V_N$ (from left to right), with $N = 2$ for (a) and $N = 3$ for (b). I assume, as for the horseshoe, uniform stretching along the strip and contraction across it.

Strips. Let $H_i = f^{-1}(V_i) \cap S$ be the horizontal strip mapped onto leg V_i (N strips). The points whose forward itinerary begins $S_1 S_2$ form a thinner horizontal strip inside H_{S_1}: there are N^2 of them (4 for (a), 9 for (b)). Going backward, the points with $S_{-1} S_0 \bullet$ form N^2 vertical strips inside the legs. Each additional symbol multiplies the number of strips by N and divides their thickness by the stretching or contraction factor.

Transition graph. Every leg V_i lies across every horizontal strip H_j, so every transition $i \to j$ is allowed: the graph is the complete directed graph with loops on N vertices (2 vertices for (a), 3 for (b)). The itineraries are all two-sided sequences over N symbols (the full N-shift).

Periodic points. Each fixed point sits in some $V_i \cap H_i$ (use Corollary 5.12: $f(H_i) = V_i$ lies across H_i). More generally, each periodic symbol string gives a periodic orbit by Corollary 5.13. The number of points with $f^n(x) = x$ is N^n, and the number of orbits of least period n is $\frac{1}{n} \sum_{d|n} \mu(n/d) N^d$:

	$n=1$	$n=2$	$n=3$	$n=4$
(a) $N=2$	2	1	2	3
(b) $N=3$	3	3	8	18

Examples: in (a), the period-2 orbit has itinerary $\overline{12}$; in (b), the period-2 orbits are $\overline{12}, \overline{13}, \overline{23}$.

Comparison with the horseshoe. Map (a) has the same symbolic structure as the horseshoe (full 2-shift, same transition graph, 2^n points of period n). The order in which the legs occur along the folded strip can differ from their left-to-right order in the square, but this changes only where the symbols are placed, not the transition graph. Map (b) is a "three-legged horseshoe": three symbols, more periodic orbits (3^n points of period n) and a larger growth rate of complexity ($\ln 3$ instead of $\ln 2$).

Answer

(a) Full 2-shift; (b) full 3-shift. In both, the transition graph is complete (every symbol can follow every symbol), the number of period-n points is N^n with $N = 2, 3$, and there are N fixed points.

Exercise 5.9

Let S be a rectangle with sides s_L, s_R, s_T, s_B. Suppose $f(s_L), f(s_R)$ surround s_L, s_R in the x-coordinate and $f(s_T), f(s_B)$ surround s_T, s_B in the y-coordinate. Show that f has a fixed point in S.

Here "surround" means that the images lie strictly outside: the x-coordinates of $f(s_L)$ lie to the left of s_L and those of $f(s_R)$ to the right of s_R, and similarly in y for the top and bottom. (If some point has $f(x) = x$ on the boundary, we are done, so assume $V(x) = f(x) - x \neq 0$ there.)

With $V = f(x) - x$ the hypotheses say:

$$\text{left side: } V_x < 0, \quad \text{right side: } V_x > 0, \quad \text{bottom: } V_y < 0, \quad \text{top: } V_y > 0.$$

Walk counterclockwise from the bottom-left corner. At the corners V lies in an open quadrant: bottom-left: III ($V_x < 0, V_y < 0$); bottom-right: IV; top-right: I; top-left: II. Along each side V stays in an open half-plane: lower half-plane on the bottom, right half-plane on the right side, upper half-plane on top, left half-plane on the left side. Inside an open half-plane the direction angle of V is a single-valued continuous function, so on each side it changes by (end angle) $-$ (start angle). On the bottom side V goes from quadrant III to IV, a strictly positive change smaller than $180°$; likewise on each of the other three sides (IV $\to$ I, I $\to$ II, II $\to$ III). So the total change is a sum of four numbers in $(0°, 180°)$, hence in $(0°, 720°)$, and it is a multiple of $360°$. Therefore it equals $360°$: the net rotation is one full turn. By Theorem 5.10, f has a fixed point in S. □

Answer

On the boundary, $V = f(x) - x$ points outward in x on the vertical sides and outward in y on the horizontal sides. Its direction makes exactly one full turn, so by the fixed-point theorem (Thm. 5.10) f has a fixed point in S.

Exercise 5.10

Let $f(x,y) = (5 - x^2 - 0.3y,\ x)$ with $V_L = [-3,-1] \times [-3,3]$ and $V_R = [1,3] \times [-3,3]$. (a) Show the net rotation of f on the boundary of each rectangle is nonzero. (b) Show $\{V_L, V_R\}$ is a Markov partition. (c) Which periods occur?

Write $V(x,y) = f(x,y) - (x,y) = \big(5 - x^2 - x - 0.3y,\ x - y\big)$.

(a) *Rectangle V_R* $(1 \le x \le 3)$. On the sides:

Side	Sign information	Corner values of V (counterclockwise)
bottom $y = -3$	$V_y = x + 3 \in [4,6] > 0$	at $(1,-3)$: (3.9, 4), angle $\approx 46^\circ$
right $x = 3$	$V_x = -7 - 0.3y < 0$	at $(3,-3)$: $(-6.1, 6)$, angle $\approx 136^\circ$
top $y = 3$	$V_y = x - 3 \le 0$	at $(3,3)$: $(-7.9, 0)$, angle 180°
left $x = 1$	$V_x = 3 - 0.3y > 0$	at $(1,3)$: $(2.1, -2)$, angle $\approx 316^\circ$

Along each side V stays in a half-plane, so the angle is determined by its corner values: $46^\circ \to 136^\circ \to 180^\circ \to 316^\circ \to 406^\circ$ $(= 46^\circ + 360^\circ)$. The net rotation is $+360^\circ$, i.e. **one counterclockwise turn** (a numerical check along the whole boundary gives exactly $+360^\circ$).

Rectangle V_L $(-3 \le x \le -1)$: similarly, V at the corners $(-3,-3), (-1,-3), (-1,3), (-3,3)$ is $(-0.1, 0)$, $(5.9, 2)$, $(4.1, -4)$, $(-1.9, -6)$, with angles 180°, 19°, -44°, -108°, and the sides stay in the half-planes $V_y \ge 0$ (bottom), $V_x > 0$ (right), $V_y < 0$ (top), $V_x < 0$ (left). The angle decreases $180^\circ \to 19^\circ \to -44^\circ \to -108^\circ \to -180^\circ$: the net rotation is -360°, **one clockwise turn**.

Both are nonzero, so f has a fixed point in each rectangle (indeed, $x = 5 - x^2 - 0.3x$ gives $x \approx 1.68$ and $x \approx -2.98$, one in each).

(b) Note $\det Df = 0.3 \ne 0$ so f is one-to-one. Now $f(x,y) = (x', y')$ with $x' = 5 - x^2 - 0.3y$ and $y' = x$: the map stretches in the x-direction and compresses in the y-direction. Take either rectangle $A = V_L$ or V_R and check that $f(A)$ lies across both V_L and V_R *horizontally*:

- The images of the top and bottom sides have $y' = x \in [-3,-1]$ or $[1,3]$, inside $[-3,3]$. ✓
- Left side of V_R $(x = 1)$: $x' = 4 - 0.3y \in [3.1, 4.9]$, to the right of both V_L and V_R. Right side of V_R $(x = 3)$: $x' = -4 - 0.3y \in [-4.9, -3.1]$, to the left of both. ✓
- Left side of V_L $(x = -3)$: $x' = -4 - 0.3y \in [-4.9, -3.1]$ (left of both). Right side of V_L $(x = -1)$: $x' = 4 - 0.3y \in [3.1, 4.9]$ (right of both). ✓

So $f(V_L)$ and $f(V_R)$ each lie across V_L and V_R, with stretching directions going to stretching directions. $\{V_L, V_R\}$ is a Markov partition.

(c) The transition graph is the complete graph on $\{V_L, V_R\}$ (all four transitions occur). By Corollary 5.13 every periodic symbol sequence gives a periodic point. The sequence $\overline{L^{k-1}R}$ has least period k and the rectangles are disjoint, so the corresponding point has least period exactly k. Hence **periodic orbits of every period $k = 1, 2, 3, \ldots$ exist.**

Answer

(a) Net rotation $+1$ turn on V_R, -1 turn on V_L (both nonzero). (b) $f(V_L)$ and $f(V_R)$ each lie across both rectangles, so $\{V_L, V_R\}$ is a Markov partition. (c) The transition graph is complete, so orbits of every period $1, 2, 3, \ldots$ exist.

Challenge 5: Computer Calculations and Shadowing

Setting. A computer makes small errors at every step, so a computed orbit is a *pseudo-orbit*: $\hat{x}_{i+1}$ is within d of $f(\hat{x}_i)$. The challenge shows that for the skinny baker map (and similar maps) a true orbit stays close to the computed one (it "shadows" it).

Step 1: a fixed point near an approximate fixed point

Let $x_0 = (a, b)$ with $|B(x_0) - x_0| < d$ in each coordinate. Let $S_0 = [a - 1.5d,\ a + 1.5d] \times [b - d,\ b + d]$ ($3d$ wide, $2d$ tall) and assume it lies on one side of the line $y = \frac{1}{2}$ (so B is linear on it and S_0 is not chopped). Say S_0 lies in the lower half; then $B(x, y) = (x/3, 2y)$ on S_0.

$B(S_0)$ *is a rectangle of width* d *and height* $4d$ centered at $c = B(x_0)$, with $|c_x - a| < d$ and $|c_y - b| < d$ by hypothesis.

- *Horizontally:* the x-extent of $B(S_0)$ is $[c_x - \frac{d}{2},\ c_x + \frac{d}{2}] \subset (a - 1.5d,\ a + 1.5d)$, since $|c_x - a| + \frac{d}{2} < 1.5d$. So the images of the left and right sides lie inside the left and right sides of S_0.
- *Vertically:* the top of $B(S_0)$ is $c_y + 2d > b - d + 2d = b + d$ (above S_0), and the bottom is $c_y - 2d < b + d - 2d = b - d$ (below S_0).

So $B(S_0)$ lies across S_0. The upper-half case is the same, with $B(x, y) = (x/3 + \frac{2}{3}, 2y - 1)$.

Fixed point. The linear branch of B is continuous on the whole plane and agrees with B on S_0. By the Colorado Corollary (Cor. 5.12) it has a fixed point in S_0, and this is a fixed point of B. Every point of S_0 is within $\sqrt{1.5^2 + 1^2}\, d = \frac{\sqrt{13}}{2} d \approx 1.80\, d < 2d$ of x_0.

Step 2: accounting for rounding error 10^{-6}

The computer evaluates B with error at most 10^{-6} per coordinate and reports $\hat{B}(x_0) = x_0$ within 10^{-6}. Then the true $B(x_0)$ differs from x_0 by at most 2×10^{-6} in each coordinate, which is below $d := 3 \times 10^{-6}$. Step 1 gives a true fixed point within $2d = 6 \times 10^{-6} < 10^{-5}$ of x_0. □

Step 3: Theorem 5.19 (periodic shadowing)

Let $x_0, \ldots, x_{k-1}, x_k = x_0$ satisfy: each coordinate of $B(x_i)$ and x_{i+1} differ by less than d. Let S_i be the $3d \times 2d$ rectangle centered at x_i (each on one side of $y = \frac{1}{2}$, as in Step 1).

The computation of Step 1 applies verbatim with x_{i+1} in place of x_0 on the target side: $B(S_i)$ is a rectangle of width d and height $4d$ centered at $B(x_i)$, which is within d of x_{i+1} in each coordinate. Hence $B(S_i)$ lies across S_{i+1} (vertically, same orientation) for $i = 0, \ldots, k - 1$, and $S_k = S_0$.

By Corollary 5.13, B^k has a fixed point $z_0 \in S_0$ whose orbit visits $S_0, \dots, S_{k-1}$: $z_i = B^i(z_0) \in S_i$. So $\{z_0, \dots, z_{k-1}\}$ is a periodic orbit with

$$|x_i - z_i| \le \tfrac{\sqrt{13}}{2} d < 2d.$$

The length k of the orbit never appears in the bound: a period-one-million orbit is shadowed just as well as a fixed point. □

Step 4: shadowing by a finite or infinite chain of rectangles

Claim. Let f be continuous and $S_0, \dots, S_k$ rectangles with $f(S_i)$ lying across S_{i+1} (same orientation). Then some $x_0 \in S_0$ has $f^i(x_0) \in S_i$ for all $0 \le i \le k$.

Proof. Put $K_j = \{x \in S_0 : f^i(x) \in S_i,\ 1 \le i \le j\}$. Each K_j is closed and bounded, hence compact, and $K_{j+1} \subset K_j$. $K_j \neq \emptyset$: because $f(S_0)$ lies across S_1 there is a curvilinear sub-rectangle $T_1 \subset S_0$ with $f(T_1) \subset S_1$ and $f(T_1)$ lying across S_1. Since $f(S_1)$ lies across S_2, the transitivity of "lying across" (Section 5.4, Figure 5.10) gives a smaller sub-rectangle $T_2 \subset T_1$ with $f^2(T_2)$ lying across S_2 and $f(T_2) \subset S_1$. Repeating, we get $T_j \subset S_0$ with $f^i(T_j) \subset S_i$ for $i \le j$, so $K_j \supset T_j \neq \emptyset$. Take $x_0 \in K_k$. □

Does k have to be finite? No. For an infinite chain $S_0, S_1, S_2, \dots$, the sets K_j form a nested sequence of nonempty compact sets, so $\bigcap_j K_j \neq \emptyset$, and any x_0 in the intersection has $f^i(x_0) \in S_i$ for *all* $i \ge 0$.

Step 5: shadowing for the baker map

Let $\{x_0, \dots, x_k\}$ satisfy: each coordinate of $B(x_i)$ and x_{i+1} differ by less than d. With S_i the $3d \times 2d$ rectangles of Step 3, $B(S_i)$ lies across S_{i+1} for $i < k$ (as in Step 3, assuming each S_i is on one side of $y = \frac{1}{2}$, so that B is continuous on it). By Step 4 there is $z_0 \in S_0$ with $B^i(z_0) \in S_i$ for all $i \le k$. The true orbit $z_i = B^i(z_0)$ satisfies $|z_i - x_i| \le \frac{\sqrt{13}}{2} d < 2d$ for $i = 0, \dots, k$. □

Step 6: other maps (the cat map)

What matters: (i) continuity (on the rectangles used), and (ii) a **uniform expanding direction and a uniform contracting direction**, with rectangles aligned to them. Then the image of a rectangle is narrower than the original in the contracting direction (so the error d is absorbed there) and taller than the original in the expanding direction (so it can span the next rectangle).

Cat map. $A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$ is symmetric with eigenvalues $e_1 = \frac{3+\sqrt{5}}{2} \approx 2.618$ (expanding) and $e_2 = \frac{3-\sqrt{5}}{2} \approx 0.382$ (contracting) and orthonormal eigenvectors u_1, u_2. Work in the plane (the lift of the cat map is the linear map A, and reducing mod 1 commutes with it). Replace the $3d \times 2d$ rectangle by a rectangle S_i centered at x_i with sides *parallel to u_2 and u_1*: width w along u_2 and height h along u_1. Since A is linear, $A(S_i)$ is a rectangle with the same orientation, of size $e_2 w \times e_1 h$, centered at Ax_i. The pseudo-orbit condition puts this center within $\sqrt{2}\, d$ (in length) of x_{i+1}, so each eigen-coordinate of the offset is less than $\sqrt{2}\, d$. $A(S_i)$ lies across S_{i+1} if

$$\frac{e_2 w}{2} + \sqrt{2}\, d \le \frac{w}{2} \iff w \ge \frac{2\sqrt{2}\, d}{1 - e_2} \approx 4.58\, d, \qquad \frac{e_1 h}{2} - \sqrt{2}\, d \ge \frac{h}{2} \iff h \ge \frac{2\sqrt{2}\, d}{e_1 - 1} \approx 1.75\, d.$$

Choose $w = 4.6d$ and $h = 1.8d$. Steps 3–5 then go through unchanged (the cat map is continuous on the torus, so there is no discontinuity line to avoid). A true orbit exists within half the diagonal, $\frac{1}{2}\sqrt{w^2 + h^2} \approx 2.47\,d < 2.5\,d$, of any d-pseudo-orbit.

Step 7: a computer plot of 10^6 cat-map iterates

Take $d = 10^{-6}$. Each plotted point is within d of the cat map image of the previous plotted point, so by Step 6 there is a true orbit within 2.5×10^{-6} of every plotted point, for all 10^6 points at once. A screen pixel is about 10^{-3} wide (for a unit square drawn ~ 1000 pixels across), which is hundreds of times larger than 2.5×10^{-6}. So yes: the dots represent a true orbit of the map to within the pixels of the screen. (It is the orbit of a slightly different starting point than the one typed in; sensitive dependence makes it impossible to do better, but it is a genuine orbit.)

Step 8: what map (5.14) lacks

Map (5.14), $f(x, y) = (x + \delta,\, y + \delta)$ with $\delta = 10^{-6}$, is a pure translation. The image of a rectangle is an identical rectangle shifted by (δ, δ): there is **no expansion and no contraction**, so no rectangle can "lie across" another and errors are never absorbed. They accumulate. For example, errors of $-\delta$ per step keep the computed orbit at $(0, 0)$ while every true orbit drifts 10^{-6} per step. The missing property is **hyperbolicity**: uniform stretching in one direction and uniform contraction in the other, which is what makes the baker map and the cat map "immune".

Summary of answers

Exercise	Result
5.1	$\prod L_k = \lvert D\rvert,\ \sum h_k = \ln \lvert D\rvert$
5.2	$h = \pm \ln \frac{1+\sqrt{5}}{2} \approx \pm 0.481$; cat-map matrix $= M^2$
5.3	A^{-1} has integer entries, so $A\mathbf{v} \equiv A\mathbf{w} \Rightarrow \mathbf{v} = \mathbf{w}$
5.4	Complete graph on $\{L, R\}$ (all four arrows) for both maps
5.5	Chaotic orbits $\leftrightarrow$ non-eventually-periodic itineraries: uncountably many
5.6	Itinerary listing every finite block (both sides) gives a dense chaotic orbit
5.7	$p(r, \theta) = (r^2, 2\theta)$; unit circle; exponents $\ln 2, \ln 2$ on it ($0, 0$ numbers inside); irrational angles chaotic
5.8	(a) full 2-shift, (b) full 3-shift; N^n points of period n
5.9	On the boundary $V = f(x) - x$ makes one full turn, so there is a fixed point
5.10	Net rotation $+1$ on V_R, -1 on V_L; Markov partition; orbits of every period
Challenge 5	$B(S_0)$ lies across S_0 ($3d \times 2d$ rectangles); fixed point within $2d$; shadowing within $2d$ for any length; cat map via eigen-aligned rectangles ($\approx 2.5d$); translation (5.14) lacks hyperbolicity